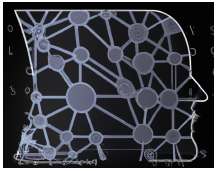
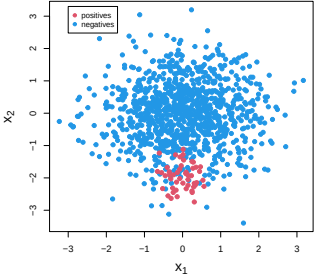
Applied Machine Learning

Dealing with Imbalanced Data



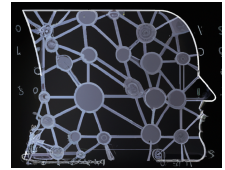


Learning goals

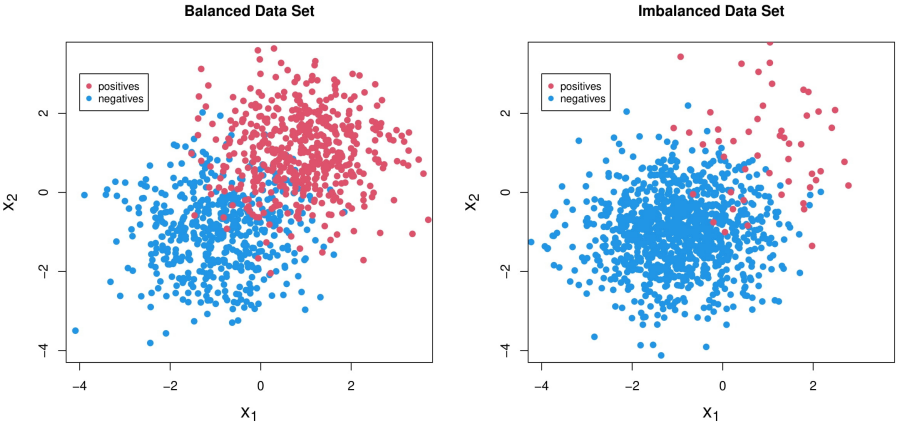
- Know what an imbalanced data set is
- Understand disadvantage of accuracy on imbalanced data
- Know techniques for handling imbalanced data sets

Introduction to Imbalanced Data

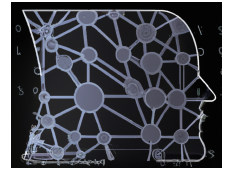
IMBALANCED DATA SETS



- Class imbalance: Ratio of classes is significantly different.
- Consequence: Undesirable predictive behavior for smaller class.
- Example: Sampling from two Gaussian distributions



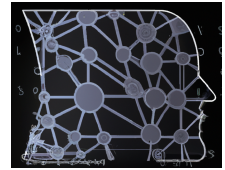
IMBALANCED DATA SETS: EXAMPLES



Domain	Task	Majority Class	Minor Class
Medicine	Predict tumor pathology	Benign	Malignant
Information retrieval	Find relevant items	Irrelevant items	Relevant items
Tracking criminals	Detect fraud emails	Non-fraud emails	Fraud emails
Weather prediction	Predict extreme weather	Normal weather	Tornado, hurricane

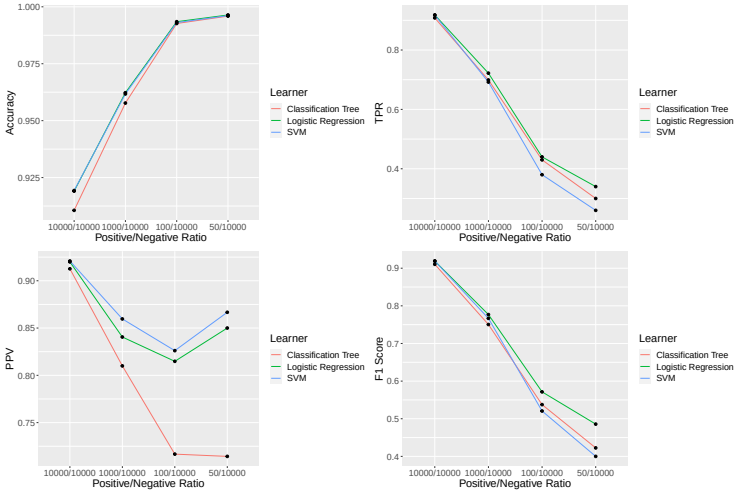
- Often, the minority class is the more important class.
- Imbalanced data can be a source of bias related to concept of fairness.

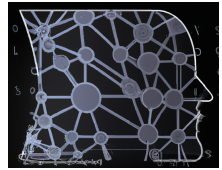
ISSUES WITH EVALUATING CLASSIFIERS



- Ideal case: correctly classify as many instances as possible
  $\Rightarrow$ High accuracy, preferably 100%.
- In practice, we often obtain on imbalanced data sets: **good** performance on the **majority** class(es), a **poor** performance on the **minority** class(es).
- Reason: the classifier is biased towards the **majority** class(es), as predicting the majority class pays off in terms of accuracy.
- Focusing only on accuracy can lead to bad performance on minority class.
- Example:
  - Assume that only 0.5% of the patients have a disease,
  - Always predicting “no disease” leads to accuracy of 99.5%

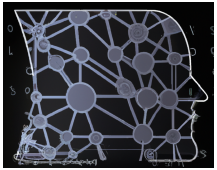
ISSUES WITH EVALUATING CLASSIFIERS





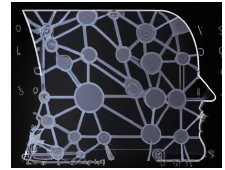
In each scenario, we have 10.000 obs in the negative class. Number of obs in positive class varies between 10.000, 1.000, 100, and 50. Train classifiers with 10-fold stratified cv. Evaluate via aggregated predictions on test set.

POSSIBLE SOLUTIONS



- Ideal performance metric: the learning is *properly* biased towards the minority class(es).
- Imbalance-aware performance metrics:
  - G-score
  - Balanced accuracy
  - Matthews Correlation Coefficient
  - Weighted macro F_1 score

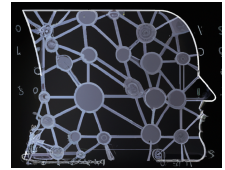
POSSIBLE SOLUTIONS



Approach	Main idea	Remark
Algorithm-level	Bias classifiers towards minority	Special knowledge about classifiers is needed
Data-level	Rebalance classes by resampling	No modification of classifiers is needed
Cost-sensitive Learning	Introduce different costs for misclassification when learning	Between algorithm- and data-level approaches
Ensemble-based	Ensemble learning plus one of three techniques above	-

Performance Measures for Imbalanced Data

RECAP: PERFORMANCE MEASURES FOR BINARY CLASSIFICATION



- We encourage readers to first go through ▸ Chapter 04.08 in I2ML.
- In binary classification ($\mathcal{Y} = \{-1, +1\}$):

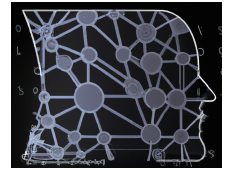
		True Class y		
		$+$	$-$	
Classification	$+$	TP	FP	$\rho_{PPV} = \frac{\#\text{TP}}{\#\text{TP}+\#\text{FP}}$
$\hat{y}$	$-$	FN	TN	$\rho_{NPV} = \frac{\#\text{TN}}{\#\text{FN}+\#\text{TN}}$
		$\rho_{TPR} = \frac{\#\text{TP}}{\#\text{TP}+\#\text{FN}}$	$\rho_{TNR} = \frac{\#\text{TN}}{\#\text{FP}+\#\text{TN}}$	$\rho_{ACC} = \frac{\#\text{TP}+\#\text{TN}}{\text{TOTAL}}$

- F_1 score balances Recall (ρ_{TPR}) and Precision (ρ_{PPV}):

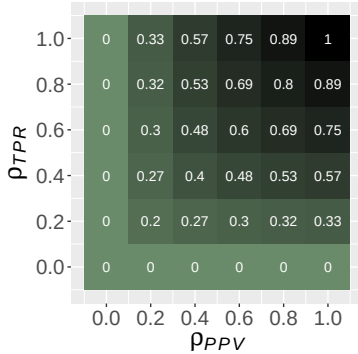
$$\rho_{F_1} = 2 \cdot \frac{\rho_{PPV} \cdot \rho_{TPR}}{\rho_{PPV} + \rho_{TPR}}$$

- Note that ρ_{F_1} does not account for TN.
- Does ρ_{F_1} suffer from data imbalance like accuracy does?

F_1 SCORE IN BINARY CLASSIFICATION

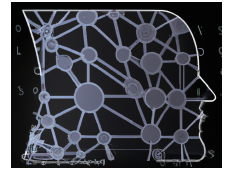


F_1 is the **harmonic mean** of ρ_{PPV} & ρ_{TPR}.
$\rightarrow$ Property of harmonic mean: tends more towards the **lower** of two combined values.



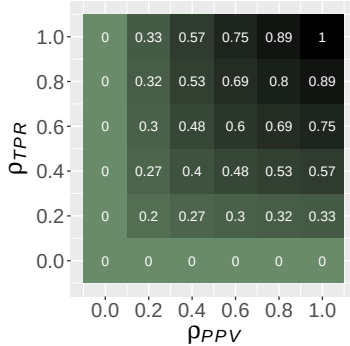
- A model with $\rho_{TPR} = 0$ or $\rho_{PPV} = 0$ has $\rho_{F_1} = 0$.
- Always predicting "negative": $\rho_{TPR} = \rho_{F_1} = 0$
- Always predicting "positive": $\rho_{TPR} = 1 \Rightarrow \rho_{F_1} = 2 \cdot \rho_{PPV}/(\rho_{PPV} + 1) = 2 \cdot n_+/(n_+ + n)$,
  $\rightsquigarrow$ small when $n_+ (= TP + FN = TP)$ is small.
- Hence, F_1 score is more robust to data imbalance than accuracy.

F_β IN BINARY CLASSIFICATION



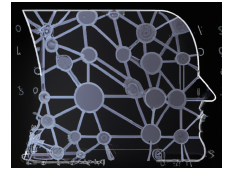
- F_1 puts equal weights to $\frac{1}{\rho_{PPV}}$ & $\frac{1}{\rho_{TPR}}$ because $F_1 = \frac{2}{\frac{1}{\rho_{PPV}} + \frac{1}{\rho_{TPR}}}$.
- F_β puts β^2 times of weight to $\frac{1}{\rho_{TPR}}$:

$$F_\beta = \frac{1}{\frac{\beta^2}{1+\beta^2} \cdot \frac{1}{\rho_{TPR}} + \frac{1}{1+\beta^2} \cdot \frac{1}{\rho_{PPV}}}$$
$$= (1+\beta^2) \cdot \frac{\rho_{PPV} \cdot \rho_{TPR}}{\beta^2 \rho_{PPV} + \rho_{TPR}}$$



- $\beta \gg 1 \rightsquigarrow F_\beta \approx \rho_{TPR}$;
- $\beta \ll 1 \rightsquigarrow F_\beta \approx \rho_{PPV}$.

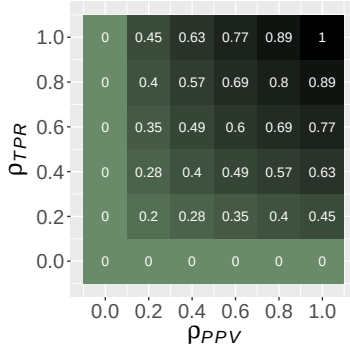
G SCORE AND G MEAN



- G score uses geometric mean:

$$\rho_G = \sqrt{\rho_{PPV} \cdot \rho_{TPR}}$$

- Geometric mean tends more towards the **lower** of the two combined values.
- Geometric mean is **larger** than harmonic mean.



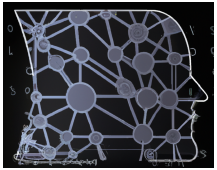
- Closely related is the G mean:

$$\rho_{Gm} = \sqrt{\rho_{TNR} \cdot \rho_{TPR}}.$$

It also considers **TN**.

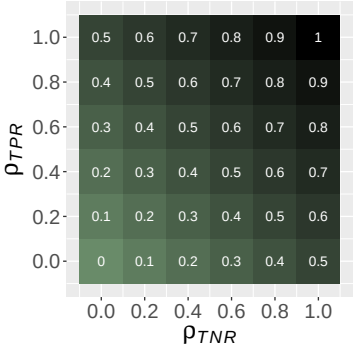
- Always predicting "negative": $\rho_G = \rho_{Gm} = 0 \rightsquigarrow$ Robust to data imbalance!

BALANCED ACCURACY



- Balanced accuracy (BAC) balances ρ_{TNR} and ρ_{TPR}:

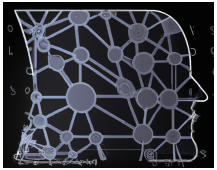
$$\rho_{BAC} = \frac{\rho_{TNR} + \rho_{TPR}}{2}$$



- If a classifier attains high accuracy on both classes or the data set is almost balanced, then $\rho_{BAC} \approx \rho_{ACC}$.

- However, if a classifier always predicts “negative” for an imbalanced data set, i.e. $n_+ \ll n_-$, then $\rho_{BAC} \ll \rho_{ACC}$. It also considers TN.

MATTHEWS CORRELATION COEFFICIENT



- Recall: Pearson correlation coefficient (PCC):

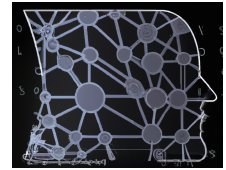
$$\mathrm{Corr}(X, Y) = \frac{\mathrm{Cov}(X, Y)}{\sigma_X \sigma_Y}$$

- View “predicted” and “true” classes as two binary random variables.
- Using entries in confusion matrix to estimate the PCC, we obtain MCC:

$$\rho_{MCC} = \frac{TP \cdot TN - FP \cdot FN}{\sqrt{(TP + FN)(TP + FP)(TN + FN)(TN + FP)}}$$

- In contrast to other metrics:
  - MCC uses all entries of the confusion matrix;
  - MCC has value in $[-1, 1]$.

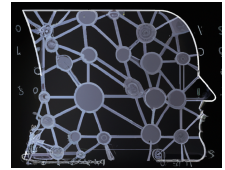
MATTHEWS CORRELATION COEFFICIENT



$$\rho_{MCC} = \frac{TP \cdot TN - FP \cdot FN}{\sqrt{(TP + FN)(TP + FP)(TN + FN)(TN + FP)}}$$

- $\rho_{MCC} \approx 1 \rightsquigarrow$ nearly zero error $\rightsquigarrow$ good classification, i.e., strong correlation between predicted and true classes.
- $\rho_{MCC} \approx 0 \rightsquigarrow$ no correlation, i.e., not better than random guessing.
- $\rho_{MCC} \approx -1 \rightsquigarrow$ reversed classification, i.e., switch labels.
- Previous measures requires defining positive class. But MCC does not depend on which class is the positive one.

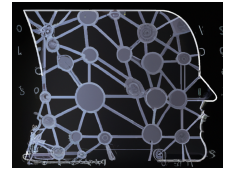
MULTICLASS CLASSIFICATION



		True Class y			
		1	2	...	g
Classification	1	n_{11} (True 1's)	n_{12} (False 1's for 2's)	...	n_{1g} (False 1's for g's)
	2	n_{21} (False 2's for 1's)	n_{22} (True 2's)	...	n_{2g} (False 2's for g's)
$\hat{y}$					
	$\vdots$	$\vdots$	$\vdots$	...	$\vdots$
	g	n_{g1} (False g's for 1's)	n_{g2} (False g's for 2's)	...	n_{gg} (True g's)

- n_{ji}: the number of i instances classified as j.
- $n_i = \sum_{j=1}^{g} n_{ji}$ the total number of i instances.
- **Class-specific** metrics:
  - True positive rate (**Recall**): $\rho_{TPR_i} = \frac{n_{ii}}{n_i}$
  - True negative rate $\rho_{TNR_i} = \frac{\sum_{j \neq i} n_{jj}}{n - n_i}$
  - Positive predictive value (**Precision**) $\rho_{PPR_j} = \frac{n_{jj}}{\sum_{i=1}^{g} n_{ji}}$.

MACRO F_1 SCORE



- Average over classes to obtain a single value:

$$\rho_{mMETRIC} = \frac{1}{g} \sum_{i=1}^{g} \rho_{METRIC_i},$$

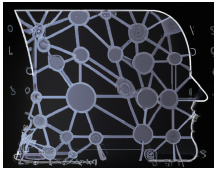
where $METRIC_i$ is a class-specific metric such as PPV_i, TPR_i of class i.

- With this, one can simply define a **macro** F_1 score:

$$\rho_{mF_1} = 2 \cdot \frac{\rho_{mPPV} \cdot \rho_{mTPR}}{\rho_{mPPV} + \rho_{mTPR}}$$

- Problem: each class equally weighted $\rightsquigarrow$ class sizes are not considered.
- How about applying different weights to the class-specific metrics?

WEIGHTED MACRO F_1 SCORE



- For imbalanced data sets, give **more weights** to **minority** classes.
- $w_1, \ldots, w_g \in [0, 1]$ such that $w_i > w_j$ iff $n_i < n_j$ and $\sum_{i=1}^{g} w_i = 1$.

$$\rho_{wmMETRIC} = \frac{1}{g} \sum_{i=1}^{g} \rho_{METRIC_i} w_i,$$

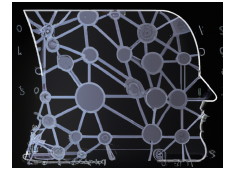
where $METRIC_i$ is a class-specific metric such as PPV_i, TPR_i of class i.

- Example: $w_i = \frac{n-n_i}{(g-1)n}$ are suitable weights.
- Weighted macro F_1 score:

$$\rho_{wmF_1} = 2 \cdot \frac{\rho_{wmPPV} \cdot \rho_{wmTPR}}{\rho_{wmPPV} + \rho_{wmTPR}}$$

- This idea gives rise to a weighted macro G score or weighted BAC.
- **Usually**, weighted F_1 score uses $w_i = n_i/n$. However, for imbalanced data sets this would **overweight** majority classes.

OTHER PERFORMANCE MEASURES



- “Micro” versions, e.g., the micro TPR is $\frac{\sum_{i=1}^{g} TP_i}{\sum_{i=1}^{g} TP_i + FN_i}$

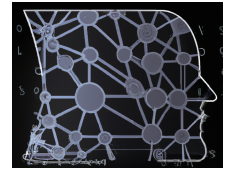
- MCC can be extended to:

$$\rho_{MCC} = \frac{n \sum_{i=1}^{g} n_{ii} - \sum_{i=1}^{g} \hat{n}_i n_i}{\sqrt{(n^2 - \sum_{i=1}^{g} \hat{n}_i^2)(n^2 - \sum_{i=1}^{g} n_i^2)}},$$

  where $\hat{n}_i = \sum_{j=1}^{g} n_{ij}$ is the total number of instances classified as i.

- Cohen’s Kappa or Cross Entropy (see Grandini et al. (2021)) treat "predicted" and "true" classes as two discrete random variables.

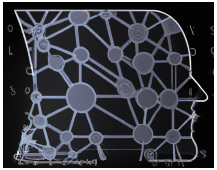
WHICH PERFORMANCE MEASURE TO USE?



- Since different measures focus on other characteristics $\rightsquigarrow$ No golden answer to this question.
- Depends on application and importance of characteristics.
- However, it is clear that accuracy usage is inappropriate if the data set is imbalanced. $\rightsquigarrow$ Use alternative metrics.
- Be careful with comparing the absolute values of the different measures, as these can be on different “scales”, e.g., MCC and BAC.
- Area under the ROC curve is also immune against class imbalance.

Cost-Sensitive Learning

COST-SENSITIVE LEARNING: IN A NUTSHELL

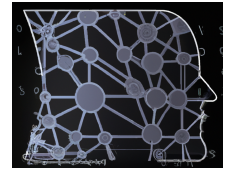


- Cost-sensitive learning:
  - Classical learning: data sets are balanced, and all errors have equal costs
  - We now assume given, unequal cost
  - And try to minimize them in expectation
- Applications:
  - Medicine — Misdiagnosing as healthy vs. having a disease
  - (Extreme) Weather prediction — Incorrectly predicting that no hurricane occurs
  - Credit granting — Lending to a risky client vs. not lending to a trustworthy client.

		Truth	
		Default	Pays Back
Pred.	Default	0	10
	Pays Back	1000	0

- In these examples, **the costs of a false negative is much higher than the costs of a false positive**.
- In some applications, the costs are **unknown** $\rightsquigarrow$ need to be specified by experts, or be learnt.

COST MATRIX

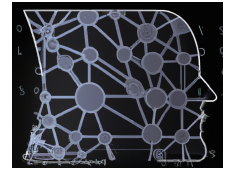


- Input: cost matrix **C**

		True Class y			
		1	2	...	g
Classification	1	$C(1,1)$	$C(1,2)$	...	$C(1,g)$
	2	$C(2,1)$	$C(2,2)$	...	$C(2,g)$
$\hat{y}$					
	$\vdots$	$\vdots$	$\vdots$	...	$\vdots$
	g	$C(g,1)$	$C(g,2)$	...	$C(g,g)$

- $C(j,k)$ is the cost of classifying class k as j,
- 0-1-loss would simply be: $C(j,k) = \mathbb{1}_{[j \neq k]}$
- **C** designed by experts with domain knowledge
  1. Too low costs: not enough change in model, still costly errors
  2. Too high costs: might never predict costly classes

COST MATRIX FOR IMBALANCED LEARNING



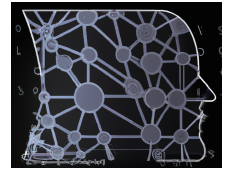
- Common heuristic for imbalanced data sets:
  - $C(j,k) = \frac{n_j}{n_k}$ with $n_k \ll n_j$,
    misclassifying a minority class k as a majority class j
  - $C(j,k) = 1$ with $n_j \ll n_k$,
    misclassifying a majority class k as a minority class j
  - 0 for a correct classification

- Imbalanced binary classification:

		True class	
		$y = 1$	$y = -1$
Pred.	$\hat{y}$ = 1	0	1
class	$\hat{y}$ = -1	$\frac{n_-}{n_+}$	0

- So: much higher costs for FNs

MINIMUM EXPECTED COST PRINCIPLE



- Suppose we have:
  - a cost matrix $\mathbf{C}$
  - knowledge of the true posterior $p(\cdot \mid \mathbf{x})$
- Predict class j with smallest expected costs when marginalizing over true classes:

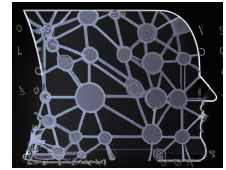
$$\mathbb{E}_{K \sim p(\cdot \mid \mathbf{x})}(C(j, K)) = \sum_{k=1}^{g} p(k \mid \mathbf{x}) C(j, k)$$

- If we trust a probabilistic classifier, we can convert its scores to labels:

$$h(\mathbf{x}) := \underset{j=1,\ldots,g}{\arg\min} \sum_{k=1}^{g} \pi_k(\mathbf{x}) C(j, k).$$

- Can be better to take a less probable class (Elkan et. al. 2001)

OPTIMAL THRESHOLD FOR BINARY CASE 1/2



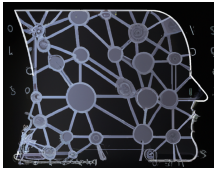
- Optimal decisions do not change if
  - **C** is multiplied by positive constant
  - **C** is added with constant shift
- Scale and shift **C** to get simpler $\mathbf{C}'$:

		True class	
		$y = 1$	$y = -1$
Pred.	$\hat{y}$ = 1	$C'(1,1)$	1
class	$\hat{y}$ = -1	$C'(-1,1)$	0

where

- $C'(-1,1) = \frac{C(-1,1)-C(-1,-1)}{C(1,-1)-C(-1,-1)}$
- $C'(1,1) = \frac{C(1,1)-C(-1,-1)}{C(1,-1)-C(-1,-1)}$

OPTIMAL THRESHOLD FOR BINARY CASE 2/2



- We predict $\mathbf{x}$ as class 1 if

$$\mathbb{E}_{K\sim p(\cdot \mid \mathbf{x})}(C'(1,K)) \leq \mathbb{E}_{K\sim p(\cdot \mid \mathbf{x})}(C'(-1,K))$$

- Let's unroll the expected value and use $\mathbf{C}'$:

$$\begin{aligned}
&p(-1 \mid \mathbf{x})C'(1,-1) + p(1 \mid \mathbf{x})C'(1,1) \leq p(-1 \mid \mathbf{x})C'(-1,-1) + p(1 \mid \mathbf{x})C'(-1,1) \\
&\Rightarrow [1 - p(1 \mid \mathbf{x})] \cdot 1 + p(1 \mid \mathbf{x})C'(1,1) \leq p(1 \mid \mathbf{x})C'(-1,1) \\
&\Rightarrow p(1 \mid \mathbf{x}) \geq \frac{1}{C'(-1,1) - C'(1,1) + 1} \\
&\Rightarrow p(1 \mid \mathbf{x}) \geq \frac{C(1,-1) - C(-1,-1)}{C(-1,1) - C(1,1) + C(1,-1) - C(-1,-1)} = c^*
\end{aligned}$$

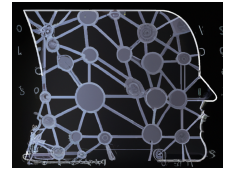
- If even $C(1,1) = C(-1,-1) = 0$, we get:

$$p(1 \mid \mathbf{x}) \geq \frac{C(1,-1)}{C(-1,1) + C(1,-1)} = c^*$$

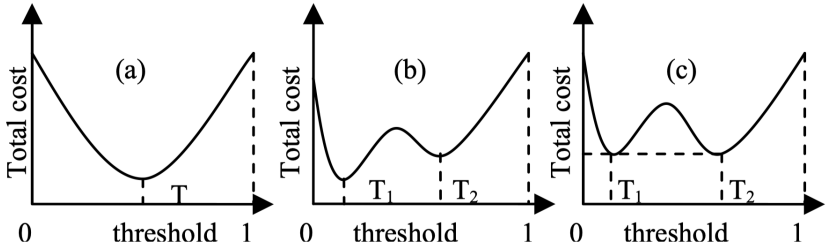
- Optimal threshold c^* for probabilistic classifier

$$h(\mathbf{x}) := 2 \cdot \mathbb{1}_{[\pi(\mathbf{x}) \geq c^*]} - 1$$

EMPIRICAL THRESHOLDING: BINARY CASE

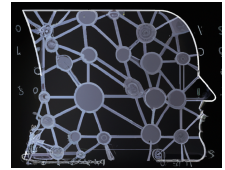


- Theoretical threshold from MECP not always best, due to e.g. wrong model class, finite data, etc.
- Simply measure costs on data with different thresholds
- Then pick best threshold (Fig.1 in Sheng et al. 2006):



- What if two equal local minima? We prefer the one with wider span
- Do this on validation data / over cross-val to avoid overfitting!

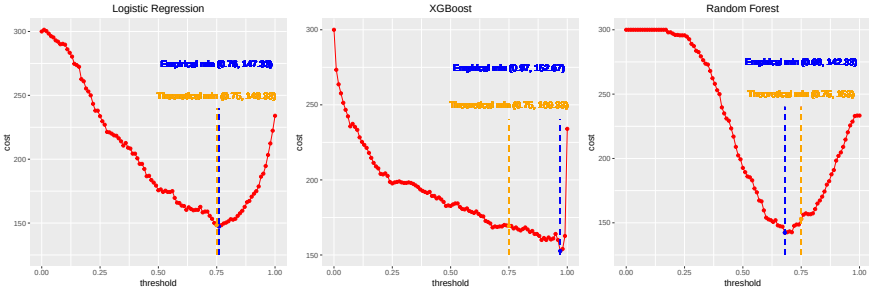
EMPIRICAL THRESHOLDING: BINARY CASE



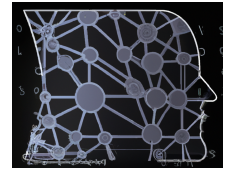
- Example: German Credit task

	True class	
	$y = \text{good}$	$y = \text{bad}$
Pred. $\hat{y} = \text{good}$	0	3
class $\hat{y} = \text{bad}$	1	0

- Theoretical: $C(good, bad)/(C(bad, good) + C(good, bad)) = 3/4 = c^*$
- Empirical version with 3-CV: For XGBoost, empirical minimum deviates substantially from theoretical version

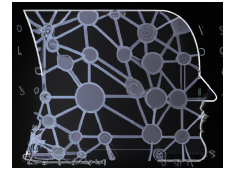


EMPIRICAL THRESHOLDING: MULTICLASS



- In the standard setting, we predict class $h(\mathbf{x}) = \arg\max_k \pi_k(\mathbf{x})$.
- Let's use g thresholds c_k now
- Re-scale scores $\mathbf{s} = (\frac{\pi(\mathbf{x})_1}{c_1}, \dots, \frac{\pi(\mathbf{x})_g}{c_g})^\top$,
- Predict class $\arg\max_k \pi_k(\mathbf{x})$.
- Compute empirical costs over cross-validation
- Optimize over g (actually: $g - 1$) dimensional threshold vector $(c_1, \dots, c_g)^T$ to produce minimal costs

BINARY INSTANCE-SPECIFIC COST LEARNING



- Assumes instance-specific costs for every observation: $\mathcal{D}^{(n)} = \{(\mathbf{x}^{(i)}, \mathbf{c}^{(i)})\}_{i=1}^{n}$, where $(\mathbf{x}^{(i)}, \mathbf{c}^{(i)}) \in \mathbb{R}^p \times \mathbb{R}^2$.
- Define "true class" as cost minimal class
- Define observation weights: $|\mathbf{c}^{(i)}[1] - \mathbf{c}^{(i)}[0]|$

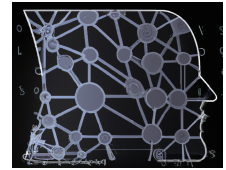
	$\mathbf{c}^{(i)}[0]$	$\mathbf{c}^{(i)}[1]$	$y^{(i)}$	$w^{(i)}$
$\mathbf{x}^{(1)}$	1	1	0	0
$\mathbf{x}^{(2)}$	1	2	0	1
$\mathbf{x}^{(3)}$	7	3	1	4

- Now solve weighted ERM:

$$\mathcal{R}_{emp}(\boldsymbol{\theta}) = \sum_{i=1}^{n} w^{(i)} L\left(y^{(i)}, f\left(\mathbf{x}^{(i)} \mid \boldsymbol{\theta}\right)\right)$$

- NB: Instances with equal costs are effectively ignored.

MULTICLASS COSTS

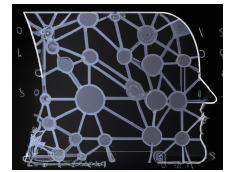


- Consider $g > 2$. Vanilla CSL is special case of instance specific, use $\mathbf{c}^{(i)}$ same for all $\mathbf{x}^{(i)}$ of the same class
- For two $\mathbf{x}^{(i)}$ with $y = 2$ and $y = 3$:

	$\mathbf{c}^{(i)}[1]$	$\mathbf{c}^{(i)}[2]$	$\mathbf{c}^{(i)}[3]$	$y^{(i)}$
$\mathbf{x}^{(1)}$	1	0	1	2
$\mathbf{x}^{(2)}$	3	1	0	3
$\mathbf{x}^{(3)}$	1	0	1	2

- Set $\mathbf{c}^{(i)}[y^{(i)}] = 0$, i.e. zero-cost for correct prediction.

CSOVO

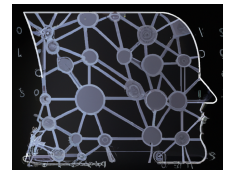


- Let $\mathcal{D}^{(n)} = \{(\mathbf{x}^{(i)}, \mathbf{c}^{(i)})\}_{i=1}^{n}$, $(\mathbf{x}^{(i)}, \mathbf{c}^{(i)}) \in \mathbb{R}^p \times \mathbb{R}^g$.
- Example:

	$\mathbf{c}^{(i)}[1]$	$\mathbf{c}^{(i)}[2]$	$\mathbf{c}^{(i)}[3]$
$\mathbf{x}^{(1)}$	0	2	3
$\mathbf{x}^{(2)}$	1	0	1
$\mathbf{x}^{(3)}$	2	0	3

- Idea: Reduction principle to binary case (weighted fit) by one-versus-one (OVO).
- For class j vs. k:
  - How to deal with the label $y^{(i)}$? $y^{(i)}$ can be neither j nor k.
  - How to deal with the costs $\mathbf{c}^{(i)}[j]$ and $\mathbf{c}^{(i)}[k]$?

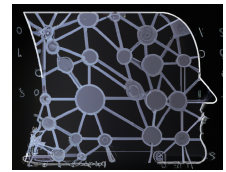
CSOVO



- When training a binary classifier $f^{(j,k)}$ for class j vs. k,
  - Choose cost min class from pair $\arg\min_{l \in \{j,k\}} \mathbf{c}^{(i)}[l]$ as ground truth
  - Sample weight is simply diff between the 2 costs $|\mathbf{c}^{(i)}[j] - \mathbf{c}^{(i)}[k]|$
- Example continued:

	$\mathbf{c}^{(i)}[1]$	$\mathbf{c}^{(i)}[2]$	$\mathbf{c}^{(i)}[3]$	$\mathbf{c}^{(i)}[1 \text{ vs } 2]$	$\tilde{y}^{(i)}[1 \text{ vs } 2]$	$w^{(i)}[1 \text{ vs } 2]$
$\mathbf{x}^{(1)}$	0	2	3	0/2	1	2
$\mathbf{x}^{(2)}$	1	0	1	1/0	2	1
$\mathbf{x}^{(3)}$	2	0	3	2/0	2	2
	$\mathbf{c}^{(i)}[1]$	$\mathbf{c}^{(i)}[2]$	$\mathbf{c}^{(i)}[3]$	$\mathbf{c}^{(i)}[2 \text{ vs } 3]$	$\tilde{y}^{(i)}[2 \text{ vs } 3]$	$w^{(i)}[2 \text{ vs } 3]$
$\mathbf{x}^{(1)}$	0	2	3	2/3	2	1
$\mathbf{x}^{(2)}$	1	0	1	0/1	2	1
$\mathbf{x}^{(3)}$	2	0	3	0/3	2	3

CSOVO



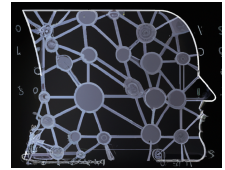
- Example continued

	$\mathbf{c}^{(i)}[1]$	$\mathbf{c}^{(i)}[2]$	$\mathbf{c}^{(i)}[3]$	$\mathbf{c}^{(i)}[1 \text{ vs } 3]$	$\tilde{y}^{(i)}[1 \text{ vs } 3]$	$w^{(i)}[1 \text{ vs } 3]$
$\mathbf{x}^{(1)}$	0	2	3	0/3	1	3
$\mathbf{x}^{(2)}$	1	0	1	-/-	-	0
$\mathbf{x}^{(3)}$	2	0	3	2/3	1	1

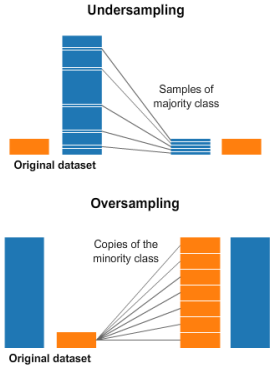
- Wrap everything up:
  1. For class j vs. k, transform all $(\mathbf{x}^{(i)}, \mathbf{c}^{(i)})$ to $(\mathbf{x}^{(i)}, \arg\min_{l \in \{j,k\}} \mathbf{c}^{(i)}[l])$ with sample-wise weight $|\mathbf{c}^{(i)}[j] - \mathbf{c}^{(i)}[k]|$.
  2. Train a weighted binary classifier $f^{(j,k)}$ using the above
  3. Repeat step 1 and 2 for different (j, k).
  4. Predict using the votes from all $f^{(j,k)}$.
- Theoretical guarantee:
  test costs of final classifier $\leq 2 \sum_{j<k}$ test cost of $f^{(j,k)}$.

Under- and Oversampling

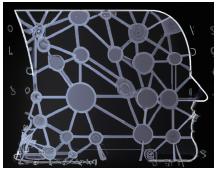
SAMPLING METHODS: OVERVIEW



- Balance training data distribution to perform better on minority classes.
- Independent of classifier $\rightsquigarrow$ very flexible and general.
- Three groups:
  - Undersampling — Removing instances of majority class(es).
  - Oversampling — Adding/Creating new instances of minority class(es). (Slower, but usually works better.)
  - Hybrid — Combining both methods.

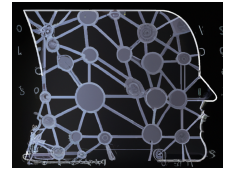


RANDOM UNDERSAMPLING/OVERSAMPLING



- Random oversampling (ROS):
  - Randomly **replicate minority** instances.
  - Prone to overfitting due to multiple tied instances.
- Random undersampling (RUS):
  - Randomly **eliminate majority** instances.
  - Might remove informative instances and destroy important concepts in data.
- Better: Introduce heuristics in removal process (RUS) and do not create exact copies (ROS).

UNDERSAMPLING: TOMEK LINKS

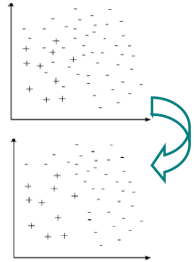


- Remove “noisy borderline” examples (very close observations of different classes) of majority class(es).
- Let $E^{(i)} = (\mathbf{x}^{(i)}, y^{(i)})$ and $E^{(j)} = (\mathbf{x}^{(j)}, y^{(j)})$ be two data points in $\mathcal{D}$.

- A pair $(E^{(i)}, E^{(j)})$ is called *Tomek link* iff there is no other data point $E^{(k)} = (\mathbf{x}^{(k)}, y^{(k)})$ such that

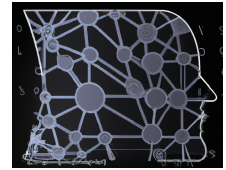
$$d(\mathbf{x}^{(i)}, \mathbf{x}^{(k)}) < d(\mathbf{x}^{(i)}, \mathbf{x}^{(j)}) \text{ or}$$
$$d(\mathbf{x}^{(j)}, \mathbf{x}^{(k)}) < d(\mathbf{x}^{(i)}, \mathbf{x}^{(j)}) \text{ holds,}$$

  where d is some distance on $\mathcal{X}$.
- $y^{(i)} \neq y^{(j)} \rightsquigarrow$ noisy borderline examples.
- Remove majority instance in each data pair in a Tomek link where $y^{(i)} \neq y^{(j)}$.



Franciso Herrera (2013), Imbalanced Classification: Common Approaches and Open Problems (URL).

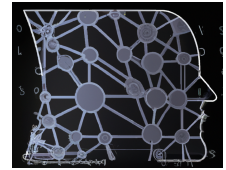
UNDERSAMPLING: OTHER APPROACHES



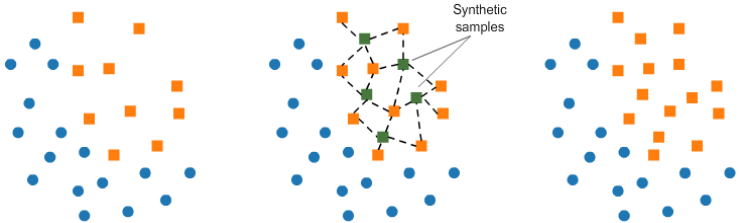
- Neighborhood cleaning rule (NCL):
  1. Find 3 nearest neighbors for each $(\mathbf{x}^{(i)}, y^{(i)})$ in $\mathcal{D}$.
  2. If $y^{(i)}$ is majority class *and* 3-NN classifies it as minority $\rightsquigarrow$ Remove $(\mathbf{x}^{(i)}, y^{(i)})$ from $\mathcal{D}$.
  3. If $y^{(i)}$ is minority class *and* 3-NN classifies it as majority $\rightsquigarrow$ Remove 3 nearest neighbors from $\mathcal{D}$.
- Condensed Nearest Neighbor (CNN): Construct a **minimally consistent** subset $\tilde{\mathcal{D}}$ of $\mathcal{D}$.
- One-sided selection (OSS): Tomek link + CNN
- CNN + Tomek link: to reduce computation of finding Tomek links $\rightsquigarrow$ first use CNN and then remove the Tomek links.
- Clustering approaches: Class Purity Maximization (CPM) and Undersampling based on Clustering (SBC).

SMOTE

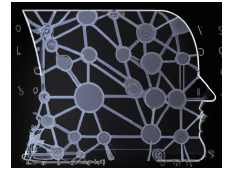
OVERSAMPLING: SMOTE



- SMOTE creates **synthetic instances** of minority class.
- Interpolate between neighboring minority instances.
- Instances are created in $\mathcal{X}$ rather than in $\mathcal{X} \times \mathcal{Y}$.
- Algorithm: For each minority class instance:
  - Find its k nearest minority neighbors.
  - Randomly select one of these neighbors.
  - Randomly generate new instances along the lines connecting the minority example and its selected neighbor.



SMOTE: GENERATING NEW EXAMPLES



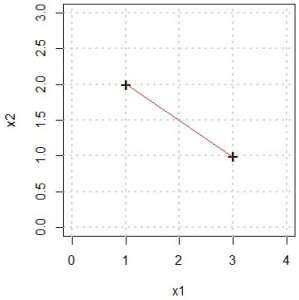
- Let $\mathbf{x}^{(i)}$ be the feature of the minority instance and let $\mathbf{x}^{(j)}$ be its nearest neighbor. The line connecting the two instances is

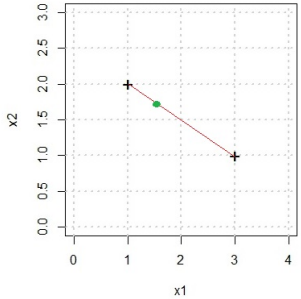
$$(1-\lambda)\mathbf{x}^{(i)} + \lambda\mathbf{x}^{(j)} = \mathbf{x}^{(i)} + \lambda(\mathbf{x}^{(j)} - \mathbf{x}^{(i)})$$

  where $\lambda \in [0, 1]$.
- By sampling a $\lambda \in [0, 1]$, say $\tilde{\lambda}$, we create a new instance

$$\tilde{\mathbf{x}}^{(i)} = \mathbf{x}^{(i)} + \tilde{\lambda}(\mathbf{x}^{(j)} - \mathbf{x}^{(i)})$$

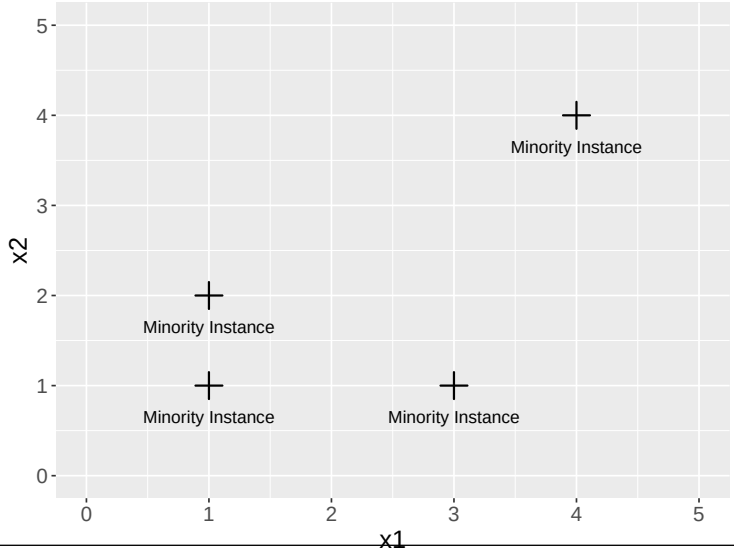
Example: Let $\mathbf{x}^{(i)} = (1, 2)^\top$ and $\mathbf{x}^{(j)} = (3, 1)^\top$. Assume $\tilde{\lambda} \approx 0.25$.





SMOTE: VISUALIZATION

For an imbalanced data situation, take four instances of the minority class. Let $K = 2$ be the number of nearest neighbors.



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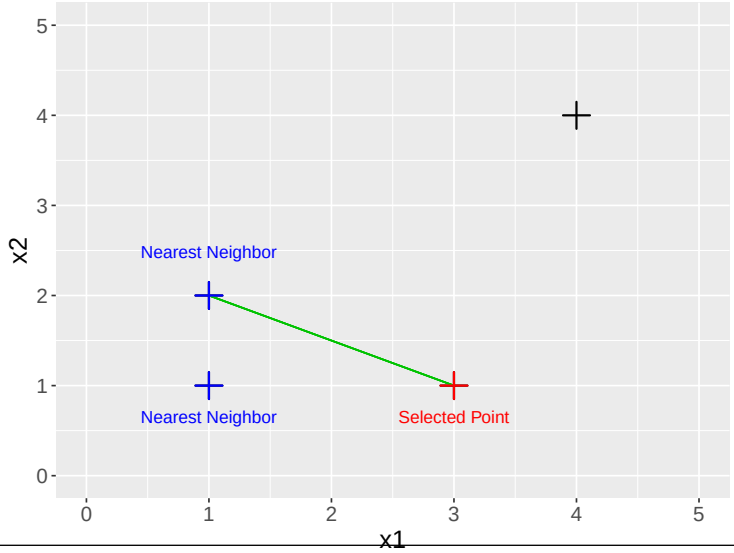

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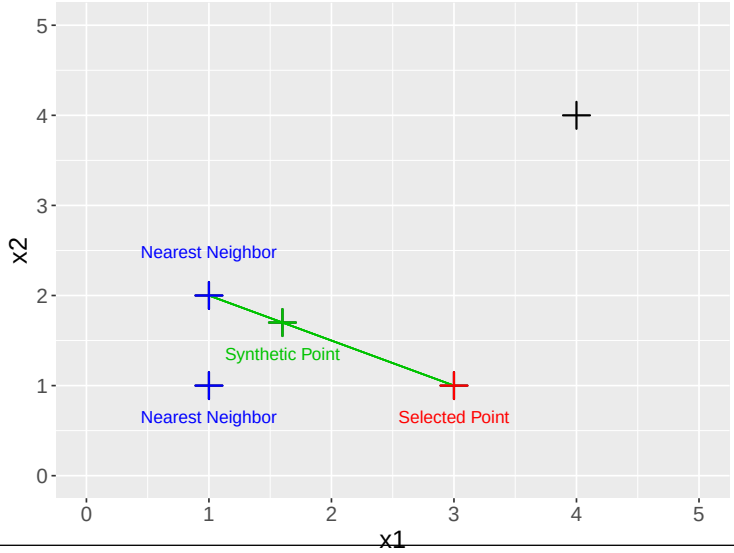
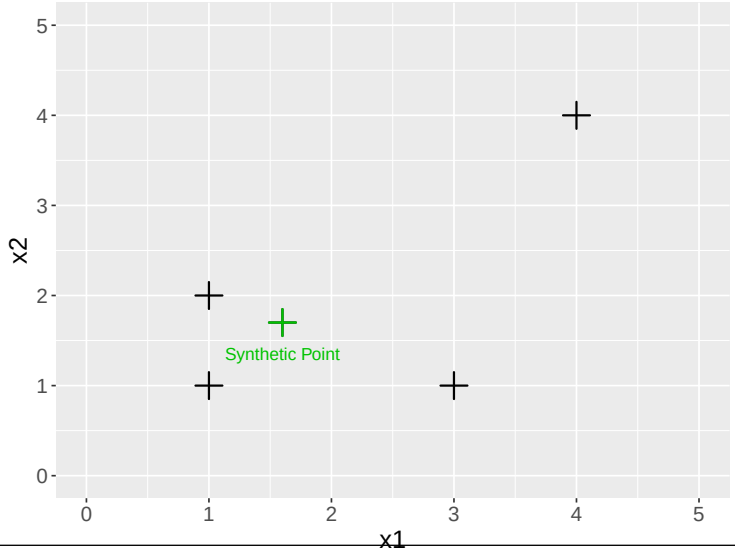


SMOTE: VISUALIZATION

For an imbalanced data situation, take four instances of the minority class. Let $K = 2$ be the number of nearest neighbors.



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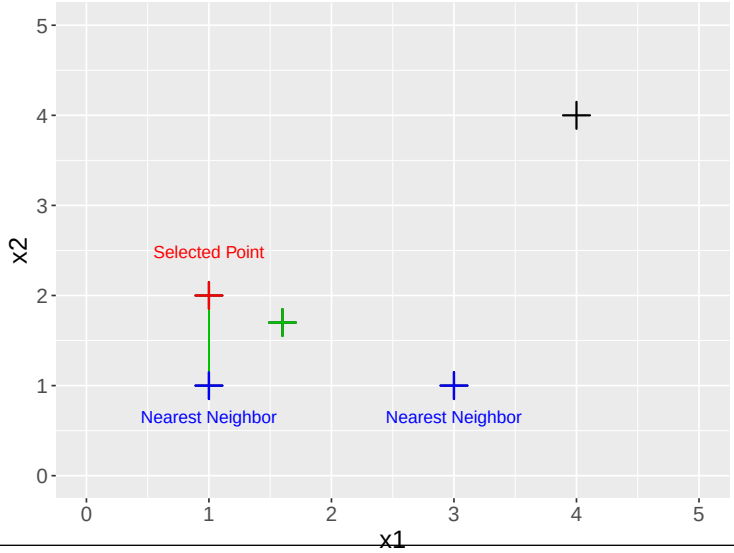

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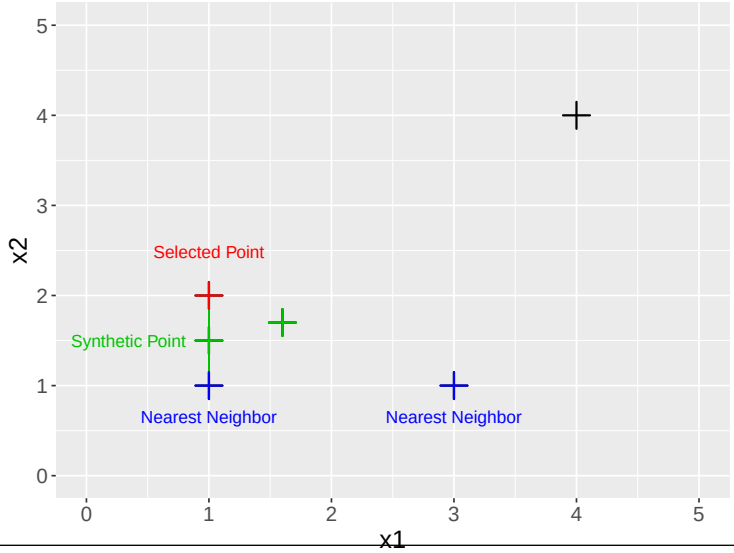

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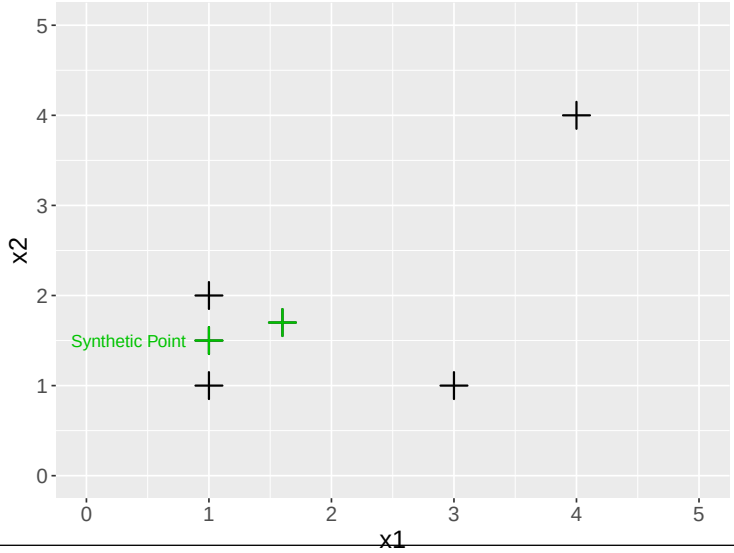

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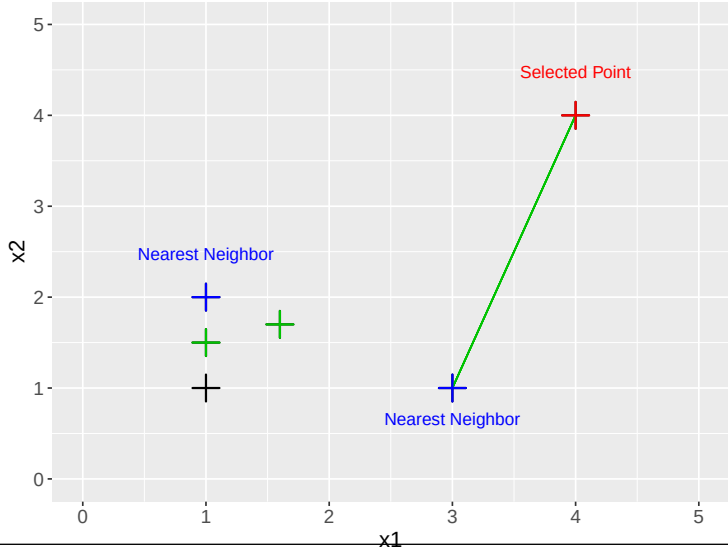

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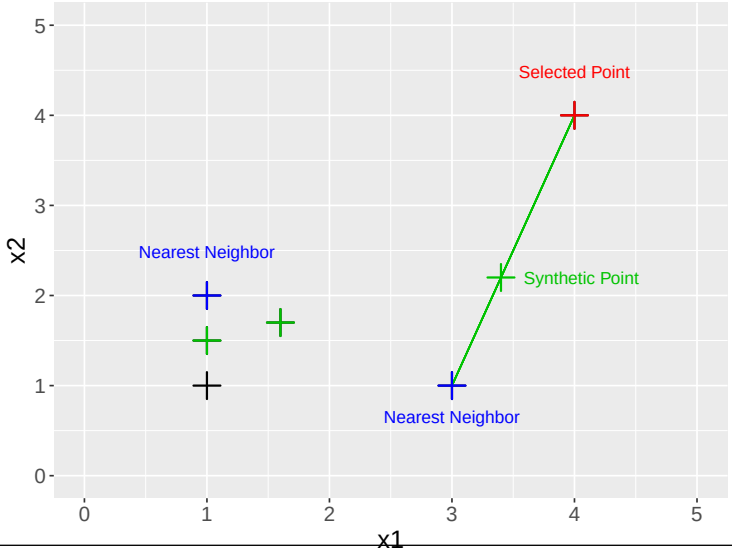

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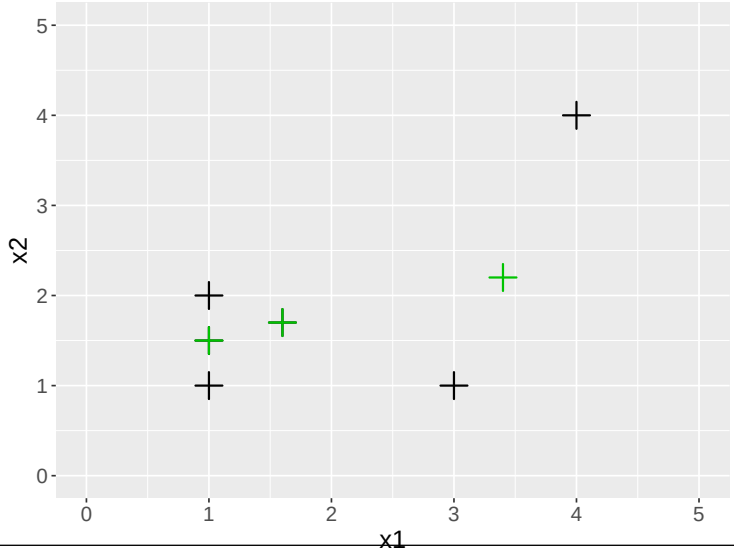
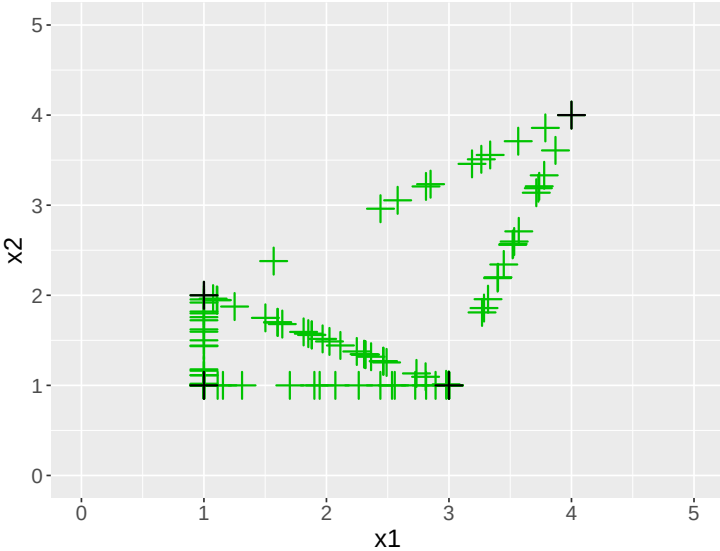


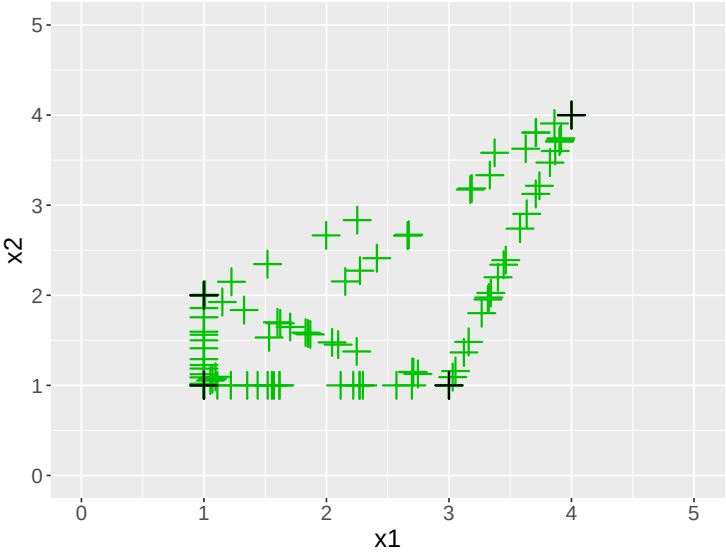
SMOTE: VISUALIZATION CONTINUED

After 100 iterations of SMOTE for $K = 2$ we get:

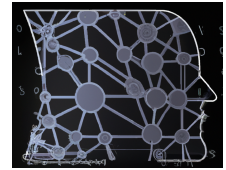


SMOTE: VISUALIZATION CONTINUED

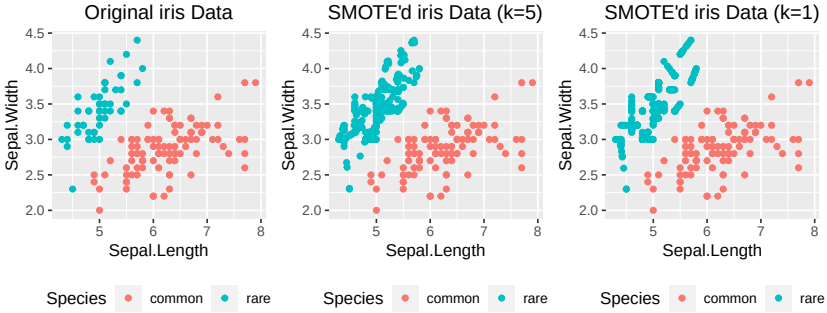
After 100 iterations of SMOTE for $K = 3$ we get:



SMOTE: EXAMPLE

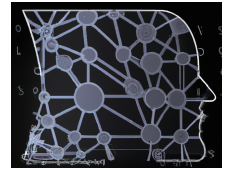


- Iris data set with 3 classes and 50 instances per class.
- Make the data set “imbalanced”:
  - relabel one class as positive
  - relabel two other classes as negative



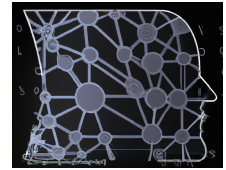
SMOTE enriches minority class feature space.

SMOTE: DIS-/ADVANTAGES



- Generalize decision region for minority class instead of making it quite specific, such as by random oversampling.
- Well-performed among the oversampling techniques and is the basis for many oversampling methods: Borderline-SMOTE, LN-SMOTE, . . . (over 90 extensions!)
- Prone to overgeneralizing as it pays no attention to majority class.

COMPARISON OF SAMPLING TECHNIQUES



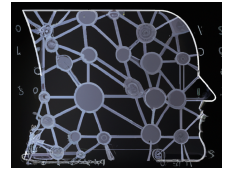
- Compare different sampling techniques on a binarized version of Optdigits dataset for optical recognition of handwritten digits.
- Use random forest with 100 trees, 5-fold cv, and F_1-Score.

Sampling technique	Class ratio	F1-Score
None	0.11	0.9239
Undersampling	0.68	0.9538
Oversampling	0.69	0.9538
SMOTE	0.79	0.9576

- Class ratios could be tuned (here done manually).
- Sampling techniques outperform base learner.
- SMOTE leads sampling techniques, although by a small margin.

Conclusion

WHEN TO COUNTERACT IMBALANCED DATA?



- Only counteract if your metric is impacted by imbalanced data
- How to counteract? Can you change to a metric that is not affected by imbalanced data?
- Check if treatment of imbalanced data has any adversarial effects REF
- Try simple methods first, especially SMOTE is highly criticized REF
- Use hyperparameter optimization to decide what method to use.
- Why not treat finding the trade-off between precision and recall as a multi-objective hyperparameter optimization problem?